

Dr. Dipankar Bhattacharya, Ph.D.

Marie Skłodowska-Curie Postdoctoral Fellow (2025–2027)

Dyson School, Imperial College London, United Kingdom



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Summary Statement

Robotics researcher working at the intersection of soft robotics, cable-driven systems, deformable-object manipulation, control, and machine learning. My research translates modelling, perception, and control methods into robotic technologies for healthcare, rehabilitation, garment manufacturing, and inspection. As a Marie Skłodowska-Curie Postdoctoral Fellow at Imperial College London, I am developing cable-driven soft reconfigurable exoskeletons for muscle-impairment rehabilitation, building on prior postdoctoral and doctoral work in robotic soft esophagus systems, cable-driven parallel robots, fabric manipulation, and learning-based control.

Education

- 2016 – 2021 ■ **Ph.D., Mechatronics Engineering**, Department of Mechanical and Mechatronics Engineering, The University of Auckland, New Zealand, 01/08/2016 – 14/04/2021.
Thesis title: *Robotic Soft Esophagus Design, Modelling, and Control for Endoprosthetic Stent Testing and Food Swallow Investigation*
- 2011 – 2013 ■ **M.Tech., Systems and Control**, Department of Electrical Engineering, Indian Institute of Technology Roorkee, India, 22/07/2011 – 04/07/2013.
Dissertation title: *Non-Linear Predictive Control Model using Relevance Vector Machine*. GPA: **8.4 / 10.0**
- 2004 – 2010 ■ **B.Tech., Electronics and Communication Engineering**, North Eastern Regional Institute of Science and Technology, India, 21/07/2004 – 23/06/2010.
Project title: *Design and Synthesis of a Sixth Order Tow Thomas Bi-quad filter*. GPA: **9.1 / 10.0**

Employment History

- 2025 – Present ■ **Marie Skłodowska-Curie Postdoctoral Fellow**, Dyson School of Design Engineering, Imperial College London, United Kingdom. Supervisor: Prof. Thrishantha Nanayakkara (Morph Lab), 01/11/2025 – Present.
CASREx: *Cable-driven Arm Soft Reconfigurable Exoskeleton for Muscle-Impairment Rehabilitation*.

Employment History (continued)

- 2025  **Senior Research Engineer–Control Systems**, Center for Transformative Garment Production (TransGP). Supervisor: Prof. Kazuhiro Kosuge, 01/05/2025 – 31/10/2025.
Physics-based cloth manipulation with policies generated from imitation learning.
- Led a multidisciplinary team to develop high-performance models for fabric pre-sewing handling and folding.
 - Investigated feasibility and oversaw development and validation of robotic systems for fabric manipulation.
 - Translated industrial requirements into research deliverables; contributed to papers, reports, and grant proposals.
 - Mentored and supervised 3 PhD students; reviewed and edited manuscripts.
- 2024 – 2025  **Postdoctoral Fellow**, Center for Transformative Garment Production (TransGP). Supervisor: Prof. Kazuhiro Kosuge, 02/07/2024 – 30/04/2025.
-  **Visiting Research Associate**, Department of Electrical and Electronic Engineering, The University of Hong Kong, 02/07/2024 – 31/10/2025.
- 2021 – 2024  **Postdoctoral Fellow**, Department of Mechanical and Automation Engineering, The Chinese University of Hong Kong. Supervisor: Prof. Darwin Lau, 01/03/2021 – 30/06/2024.
Project 1: Modelling and control of cable-driven robots with cable wrapping around obstacles and links.
Project 2: Design of a cable-driven growing vine robot for safer false-ceiling inspections.
Primary responsibilities:
- Conducted independent and collaborative research, contributing to larger projects under PI supervision.
 - Wrote research articles, proposals, and technical reports.
 - Mentored undergraduate and postgraduate students; delivered lectures for **MAEG5090 Topics in Robotics** (class size: 50 students).
- 2021 – 2022  **Visiting Postdoctoral Fellow**, Laboratoire des Sciences du Numérique de Nantes (LS2N), École Centrale de Nantes. Supervisors: Prof. Stephane Caro and Prof. Darwin Lau, 26/10/2021 – 05/01/2022.
Project: Iterative-learning control of cable-driven parallel robots.
- 2017 – 2020  **Graduate Teaching Assistant** at The University of Auckland.
Courses taken: MECHENG 706: Mechatronics Design Projects, MECHENG 306: Design of Sensing and Actuating Systems, ENGGEN 115: Principles of Engineering Design, ENGGEN 131: Introduction to Engineering Computation and Software Development.
- Supervised undergraduate students on design projects, guiding problem-solving and ensuring practical solutions.
 - Delivered tutorials, workshops, and drop-in clinics; provided support via online forums (Piazza, Canvas).
 - Managed laboratory sessions and coordinated TA activities, including safety briefings and student progress monitoring.

Employment History (continued)

- 2018 – 2019  **Research Assistant** at The University of Auckland, under the direction of Prof Peter Xu, 09/01/2018 – 08/01/2019.
- 2013–2016  **Lecturer (Assistant Professor Grade-1)**, School of Electrical, Electronics and Communication Engineering, Galgotias University, India, 19/07/2013–18/06/2016.
- Delivered lectures for undergraduate and postgraduate courses including Control Systems, Process Control, and Linear Integrated Circuits.
 - Developed teaching materials for lectures and laboratory sessions.
 - Supported students in a pastoral and advisory role.
 - Managed laboratory sessions and supervised staff and lab technicians.
 - Supervised final-year undergraduate projects.

Research Publications

Journal (Published/Accepted)

- 1 Jin*, B., Kobayashi*, A., Bhattacharya, D., Seino, A., Tokuda, F., Tien, N., & Kosuge, K. (2025). Automated Straight-line Sewing of Stretchable Fabrics with Different Lengths [*Equal contribution.]. *IEEE Robotics and Automation Letters*, 1–8.  <https://doi.org/10.1109/LRA.2025.3619791>
- 2 Bhattacharya, D., Cheung, T. K., Wang, Y., & Lau, D. (2025). Kinematic and Dynamic Modeling of Cable-Object Interference and Wrapping in Complex Geometrical-Shaped Cable-Driven Parallel Robots. *Mechanism and Machine Theory*, 214, 106092.  <https://doi.org/10.1016/j.mechmachtheory.2025.106092>
- 3 Bhattacharya, D., Chan, Y. P., Shang, S., Chan, Y. S., Tan, Y., & Lau, D. (2023). Tri-Space Operational Control of Redundant Multilink and Hybrid Cable-Driven Parallel Robots Using an Iterative-Learning based Reactive Approach. *IEEE Transactions on Control Systems and Technology*.  <https://doi.org/10.1109/TCST.2023.3263689>
- 4 Bhattacharya, D., Cheng, L. K., & Xu, W. (2021). Nonlinear Model Predictive Control of a Robotic Soft Esophagus for Endoprosthetic Stent Testing. *IEEE Transactions on Industrial Electronics*.  <https://doi.org/10.1109/TIE.2021.3121755>
- 5 Bhattacharya, D., Jesna, S., Cheng, L. K., & Xu, W. (2021). RoSE: A Robotic Soft Esophagus for Endoprosthetic Stent Testing. *Soft Robotics*.  <https://doi.org/10.1089/soro.2019.0205>
- 6 Dang, Y., Liu, Y., Hashem, R., Bhattacharya, D., Allen, J., Stommel, M., Cheng, L. K., & Xu, W. (2021). SoGut: A Soft Robotic Gastric Simulator. *Soft Robotics*.  <https://doi.org/10.1089/soro.2019.0136>
- 7 Bhattacharya, D., Cheng, L. K., & Xu, W. (2020). Sparse Machine Learning Discovery of Dynamic Differential Equation of an Esophageal Swallowing Robot. *IEEE Transactions on Industrial Electronics*, 67(6), 4711–4720.  <https://doi.org/10.1109/TIE.2019.2928239>
- 8 Bhattacharya, D., Nisha, M. G., & Pillai, G. (2015). Relevance vector-machine-based solar cell model. *International Journal of Sustainable Energy*, 34(10), 685–692.  <https://doi.org/10.1080/14786451.2014.885030>

Books and Chapters

- 1 Bhattacharya, D., Cheng, L. K., Dirven, S., & Xu, W. (2019). *Artificial intelligence approach to the trajectory generation and dynamics of a soft robotic swallowing simulator*. Springer International Publishing.  https://doi.org/10.1007/978-3-319-78452-6_1

Conference Proceedings

- 1 Dong, W., Bhattacharya, D., Kobayashi, A., Seino, A., Tokuda, F., Huang, X., Tang, K., Tien, N. C., & Kosuge, K. (2025). Precise Top-layer Fabric Segmentation for Fabric Destacking with Edge- and Shape-Aware Deep Networks. *2025 IEEE International Conference on Mechatronics and Automation (ICMA)*, 1343–1348.
- 2 Bhattacharya, D., Cheng, L. K., & Xu, W. (2018). Testing and Analysis of Migration Displacement of a Flared Stent Deployed in an Esophageal Swallowing Robot. *25th International Conference on Mechatronics and Machine Vision in Practice (M2VIP)*, 1–6. <https://doi.org/10.1109/M2VIP.2018.8600911>
- 3 Bhattacharya, D., Cheng, L. K., Dirven, S., & Xu, W. (2017). Actuation planning and modeling of a soft swallowing robot. *24th International Conference on Mechatronics and Machine Vision in Practice (M2VIP)*, 1–6. <https://doi.org/10.1109/M2VIP.2017.8211476>

In Review/Preparation

- 1 Bhattacharya*, D., Tang*, K., Hang, X., Tokuda, F., Tien, N., & Kosuge, K. (2026). RTFF: Random-to-Target Fabric Flattening Policy Using Dual-Arm Manipulator, *IEEE International Conference on Robotics and Automation (ICRA)* [in review. *Equal contribution].
- 2 Wang, W., Bhattacharya, D., Kobayashi, A., Seino, A., Tokuda, F., Tien, N., & Kosuge, K. (n.d.). Automated Fabric Alignment Using Global-Local Optimization and Point Cloud Registration, *IEEE Transactions on Automation Science and Engineering (T-ASE)* [in review].
- 3 Cheung, T., Bhattacharya, D., & Lau, D. (n.d.). Rapidly Deployable Deformable Frame Cable-Driven Parallel Robots with Small Footprint [in preparation].
- 4 Wang, Y., Bhattacharya, D., & Lau, D. (n.d.). Static modelling and workspace analysis with arbitrary cable attachment design in a cable-driven parallel robot with passive backbone, *IEEE Transactions on Automation Science and Engineering* [in preparation].

Grants and Scholarships

Grants/Fellowships

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| 2025 | Recipient, Horizon Europe Marie Skłodowska-Curie Postdoctoral Fellowship for the project “CAS-REx” (Grant No. 101205678), hosted at Imperial College London . Funded under HORIZON-MSCA-2024-PF-01-01. Duration: 24 months; value: EUR 276,187.92 . |
| 2023 | Co-authored Innovation and Technology Fund (ITF), Partnership Research Programme (PRP) grant. |
| 2022 | Co-authored Research Talent Hub for ITF project (RTH-ITF) grant. |
| 2018 | Recipient, Dynamic Testing of Implant Stents Using Robotic Soft Esophagus (RoSE), BvW Holding, Cham, Switzerland (NZD 39,000; 1 year). |
| 2018, 2019 | Recipient, Riddet Travel Grant, Riddet Institute Centre of Research Excellence (NZD 3,000). |

Scholarships

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| 2019–2020 | Recipient, Doctor of Philosophy Scholarship, University of Auckland. |
| 2016–2019 | Recipient, Doctor of Philosophy Scholarship, Riddet Institute Centre of Research Excellence. |
| 2011–2013 | Recipient, Master of Technology Scholarship, Ministry of Human Resource Development (MHRD), Government of India. |

Academic Awards and Professional Service

Academic Awards

- 2025  Selected, **European Talent Academy 2026** (Imperial College London, TU Munich, Politecnico di Milano).
- 2017  Recipient, **Best Conference Paper Award**, 5th International Conference on Robotic Intelligence Technology and Applications (RiTA), Daejeon, Korea.
-  **Ranked 2nd** in poster presentation, Delivery of Functionality in Complex Food Systems (DoF 2017) Conference.

Competitive Examinations

- 2011  **Percentile: 99.12% (Rank 1,211/137,853)** in Graduate Aptitude Test in Engineering (GATE).
- 2010  **Percentile: 96.00%** in Graduate Aptitude Test of Engineering (GATE).
- 2004  **Ranked 31st** in All India Junior Mathematics Olympiad.

Professional Activities

- 05/02/26  Invited Speaker, **Dyson Engineering Departmental Seminar**, Imperial College London.
- 29/01/26  Invited Speaker, **Imperial Robotics Postgrad Committee**, Imperial College London. Talk: *Learning-based Fabric Manipulation with Mesh Perception and Visual Servoing*.
- 28/01/26  Delivered invited lecture on **Robotics Simulation and Control using PyDrake** for DESE71002 Robotics Research Projects, Dyson School of Design Engineering, Imperial College London.
- 08/08/25  Invited Speaker, **World Robotics Conference (WRC 2025)**, Beijing, China. Talk: *Fabric Modelling and Perception for Garment Manufacturing*.
- 03/08/25  Workshop Organizer and Presenter, **IEEE ICMA 2025**, Beijing, China. Talk: *Mesh-based Fabric State Estimation and Manipulation*.
- 29/05/23  Poster Presentation, **IEEE ICRA 2023**, ExCeL London, United Kingdom.
- 2018  Presenter, **BioMouth 2018 Meet**, Auckland, New Zealand.
- 2021 – Present  Reviewer for *IEEE Robotics and Automation Letters* (RA-L), *IEEE International Conference on Robotics and Automation* (ICRA), and *ASME Journal of Mechanisms and Robotics*.
- 2017 – Present  Member, Institute of Electrical and Electronics Engineers (IEEE).
- 2014  Former member of the Electrical Engineering Society (EES), IIT Roorkee, and NERIST Electronics Society (NES), NERIST.
- 2006  Awarded an Outstanding Certificate by the National Service Scheme (NSS) for contributions to blood donation camps, village cleanliness programmes, and children's education.

Skills

- Languages  English and Hindi: professional reading, writing, and speaking proficiency.
- Research areas  Soft robotics, cable-driven robots, deformable-object manipulation, robot modelling and control, visual servoing, learning-based control, and robotic systems for healthcare and manufacturing.
- Programming  Python, MATLAB, PyTorch, Gym framework, ROS, LabVIEW, assembly-level programming (8085, 8086, MASM32, and 8051), Git, and LaTeX.
- Software  Blender, SOLIDWORKS, Autodesk Inventor, Inkscape, and Lightroom.
- Hardware  Raspberry Pi, myRIO, and Arduino.